

Demand

JEB104 Microeconomics I
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Demand

Optimal bundle: $\mathbf{x}$ demanded under given prices $\mathbf{p}$, income m , preferences $\succeq$

(Marshallian/Walrasian) Demand for good i :

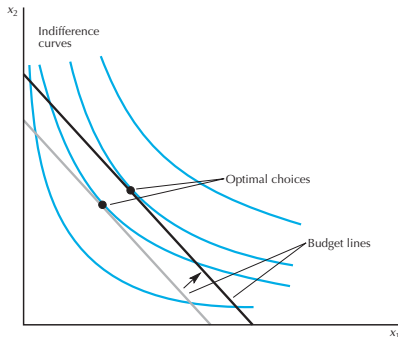
$$x_i^*(p_1, \dots, p_n, m) = x_i^*(\mathbf{p}, m) = D_i(\mathbf{p}, m)$$

Characteristics of $x(\mathbf{p}, m) = D(\mathbf{p}, m)$:

- 1 Homogeneity of degree zero:
 $x(\alpha \mathbf{p}, \alpha m) = x(\mathbf{p}, m)$ for any $\mathbf{p}$, m and $\alpha > 0$
- 2 Walras' law:
 $\mathbf{p} \cdot \mathbf{x} = m$ for any $\mathbf{x} \in x(\mathbf{p}, m)$
- 3 Uniqueness:
If $u(\mathbf{x})$ is strictly quasiconcave, then $x(\mathbf{p}, m)$ consists of a single element.
(unique solution $x^*(\mathbf{p}, m)$ for any combination $(\mathbf{p}, m)$)

Income Changes: Normal Goods

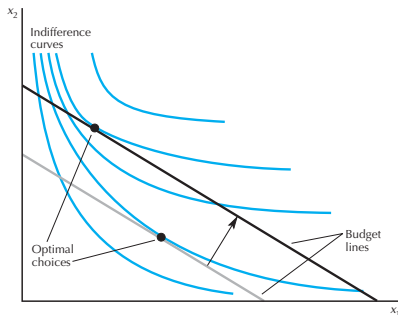
Prices fixed, change in income $\rightarrow$ change in demand



Increase in income $\rightarrow$ more of all goods

Normal good: $\frac{\Delta x}{\Delta m} > 0$

Income Changes: Inferior Goods



Increase in income $\rightarrow$ more of some goods and less of some goods

Inferior good: $\frac{\Delta x}{\Delta m} < 0$

Above: x_2 normal good, x_1 inferior good

Inferiority depends on the level of income.

Can all goods be inferior? Never.

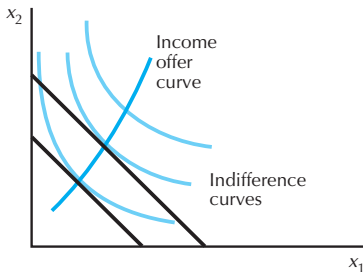
Income Offer Curve and Engel Curve

Income offer curve = income expansion path

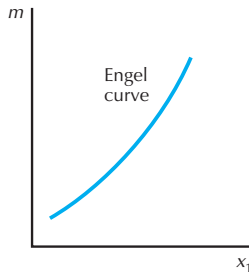
- bundles demanded at the different levels of income

Engel curve (EC)

- demand for a good i as a function of income, all prices being fixed, $D_i(\bar{p}, m)$



A Income offer curve



B Engel curve

Engel Curve

Slope of Engel curve $\frac{\partial D_i(\bar{p}, m)}{\partial m}$ = marginal propensity to consume MPC_i

EC increasing: $\frac{\partial D_i(\bar{p}, m)}{\partial m} = MPC_i > 0 \leftrightarrow$ normal good

EC decreasing: $\frac{\partial D_i(\bar{p}, m)}{\partial m} = MPC_i < 0 \leftrightarrow$ inferior good

If $\frac{\partial D_i(\bar{p}, m)}{\partial m} = MPC_i = 0 \leftrightarrow$ normal good becomes inferior

Average propensity to consume $APC_i = \frac{D_i(\bar{p}, m)}{m}$

$p_i \cdot APC_i = \frac{p_i x_i}{m}$ = share of expenditures on x_i in consumer's income m

Income Elasticity of Demand

$$e_m^i(p, m) = \frac{\% \Delta x_i}{\% \Delta m} = \frac{\frac{\partial D_i(p, m)}{\partial m}}{\frac{D_i(p, m)}{m}} = \frac{MPC_i}{APC_i}$$

$$e_m^i(p, m) = \frac{p_i}{p_i} \cdot \frac{\frac{\partial D_i(p, m)}{\partial m}}{\frac{D_i(p, m)}{m}} = \frac{\frac{p_i \Delta x_i}{\Delta m}}{\frac{p_i x_i}{m}} = \frac{\frac{p_i \Delta x_i}{\Delta m}}{\frac{p_i x_i}{m}}$$

Normal good $e_m^i(p, m) > 0$:

- ① $e_m^i(p, m) > 1$: demand is **elastic** with respect to income (luxury goods)
- ② $e_m^i(p, m) < 1$: demand is **inelastic** with respect to income (necessities)
- ③ $e_m^i(p, m) = 1$: demand is **unitary elastic** with respect to income
- ④ $e_m^i(p, m) = \infty$: demand is **perfectly elastic** with respect to income
- ⑤ $e_m^i(p, m) = 0$: demand is **perfectly inelastic** with respect to income

Inferior good $e_m^i(p, m) < 0$

Why Is Income Elasticity of Demand Useful?

A firm would like to know income elasticity of demand for their products.

- Sales forecasting
- Pricing policy
- Diversification
 - Demand for normal goods pro-cyclical
 - Demand for inferior goods counter-cyclical

Income elasticities: Examples

Janda, Mikolasek and Netuka (2010): Complete almost ideal demand system approach to the Czech alcohol demand. *Agriculture Economics*, 56(9), 421–434.

- Data from the Czech Budget Household Survey, around 11,000 observations
- Estimates: industrial and manufactured goods 1.34, services 0.98, food 0.60
- Food group: animal products 0.95, vegetables 0.99, cereals and other food 0.92, drinks 1.22
- Drinks group: beer 1.33, wine 0.76, non-alcoholic drinks 1.06

Havranek and Kokes (2015): Income elasticity of gasoline demand: A meta-analysis. *Energy Economics*, 47, 77-86

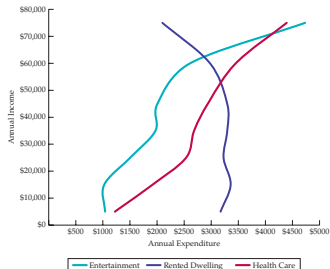
- 240 papers
- Mean elasticities: short-run 0.28, long-run 0.66
- Estimates: short-run 0.10, long-run 0.23

Consumer expenditure in the United States

TABLE 4.1 ANNUAL U.S. HOUSEHOLD CONSUMER EXPENDITURES

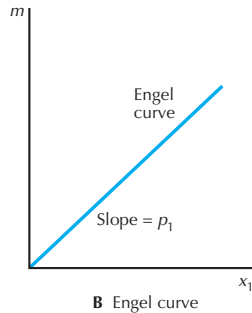
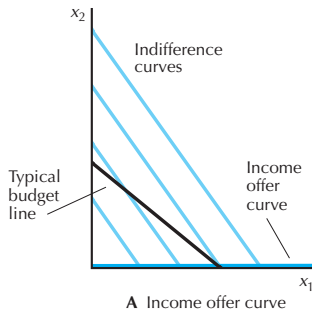
EXPENDITURES (\$ ON:	INCOME GROUP (2009 \$)						
	LESS THAN \$10,000	10,000–19,999	20,000–29,999	30,000–39,999	40,000–49,999	50,000–69,999	70,000 AND ABOVE
Entertainment	1,041	1,025	1,504	1,970	2,008	2,611	4,733
Owned Dwelling	1,880	2,083	3,117	4,038	4,847	6,473	12,306
Rented Dwelling	3,172	3,359	3,228	3,296	3,295	2,977	2,098
Health Care	1,222	1,917	2,536	2,684	2,937	3,454	4,393
Food	3,429	3,529	4,415	4,737	5,384	6,420	9,761
Clothing	799	927	1,080	1,225	1,336	1,608	2,850

Source: U. S. Department of Labor, Bureau of Labor Statistics, "Consumer Expenditure Survey, Annual Report 2010."



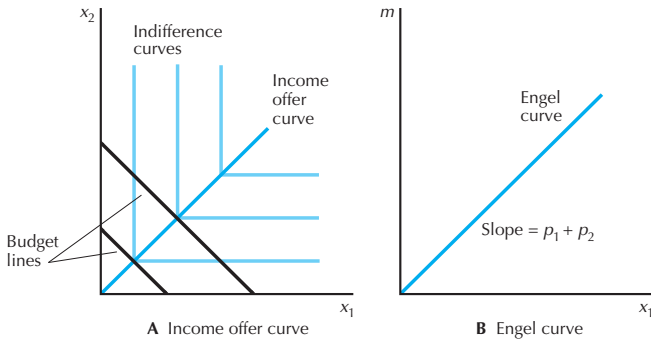
Engel Curve: Examples

Perfect substitutes



Engel Curve: Examples

Perfect complements

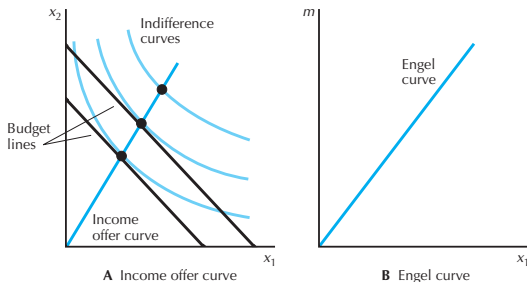


Perfect substitutes and perfect complements = example of homothetic preferences

Engel Curve: Examples

Homothetic preferences

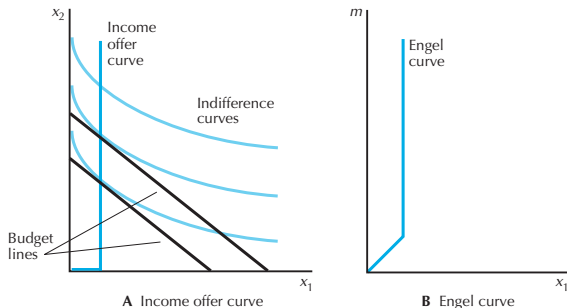
- for any two bundles x, y : if $x \sim y$, then $ax \sim ay$ for any $a > 0$
- MRS constant along any ray from the origin
- for any level of m , consumer will demand goods in the same proportion
- $\uparrow$ in income $\rightarrow$ proportional increase in goods purchased
- income offer curve and Engel curve are straight lines



Homothetic preferences may be represented by a homogeneous utility function.

Engel Curve: Examples

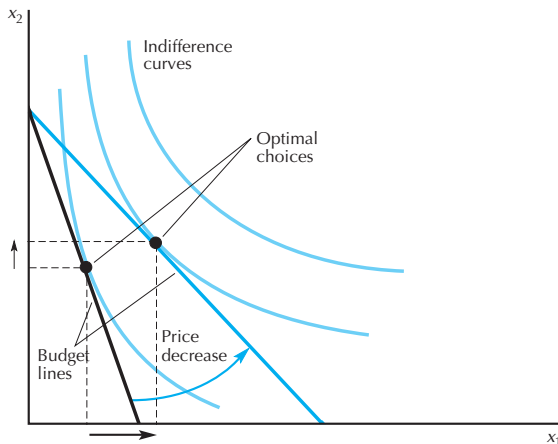
Quasilinear preferences



Income beyond the given level changed $\rightarrow$ demand for x_1 unchanged

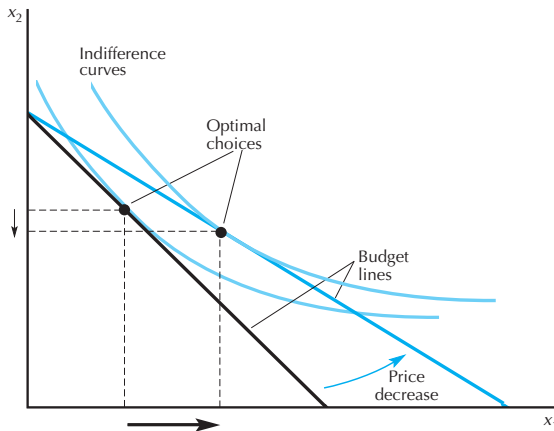
Price Change

Income fixed, change in price $\rightarrow$ change in demand



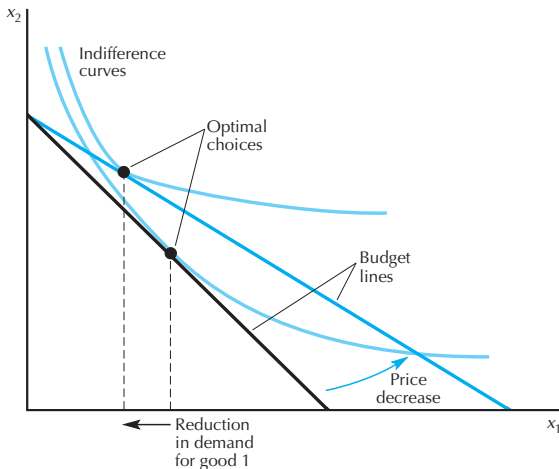
1. $\downarrow$ in price of good $i \rightarrow \uparrow$ in demand for all goods

Price Change



2. $\downarrow$ in price of good $i \rightarrow \uparrow$ in demand for good i and $\downarrow$ (or no change) in demand for other goods

Price Change



3. $\downarrow$ in price of good $i \rightarrow \downarrow$ in demand for good i and $\uparrow$ in demand at least for one other good

Price Change

Own-price change:

- ① Ordinary good: $\frac{\Delta x_i(p, m)}{\Delta p_i} < 0$
- ② Giffen good: $\frac{\Delta x_i(p, m)}{\Delta p_i} > 0$

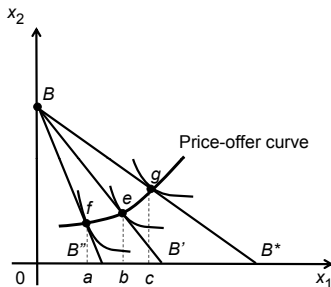
Cross-price change:

- ① x_i is gross substitute of x_j : $\frac{\Delta x_i(p, m)}{\Delta p_j} > 0$
- ② x_i is gross complement of x_j : $\frac{\Delta x_i(p, m)}{\Delta p_j} < 0$

Price Offer Curve

Price offer curve = price-consumption path

- connected optimal consumption bundles as price of one good varies



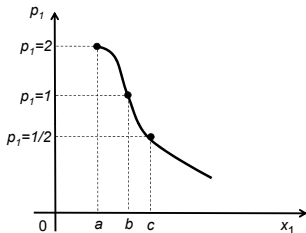
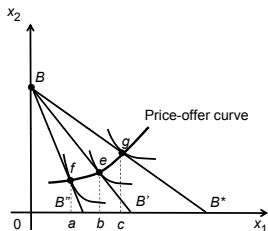
Positive slope:

x_1 ordinary and x_2 is its gross complement

Negative slope:

1. x_1 ordinary and x_2 is its gross substitute
2. x_1 Giffen and x_2 is its gross complement

Demand Curve



Demand curve for x_i : $D_i(p, m) = D_i(p_i)$

- relationship between quantity demanded and price as the price varies (m and prices of other goods being constant)
- image of the price-consumption path
- Inverse demand function = price as function of quantity
- Downward-sloping for ordinary good:

$$\frac{\partial D_i(p, m)}{\partial p_i} < 0$$
- Upward-sloping for Giffen good:

$$\frac{\partial D_i(p, m)}{\partial p_i} > 0$$
- *Look at examples of price offer curves and demands in the textbook.*

Price Elasticity of Demand

$$e_{p_i}^i = \frac{\% \Delta x_i}{\% \Delta p_i} = \frac{\frac{\partial D_i(p, m)}{\partial p_i}}{\frac{D_i(p, m)}{p_i}}$$

Ordinary good $e_{p_i}^i < 0$, Giffen good $e_{p_i}^i > 0$

Price Elasticity of Demand: Impact on Expenditure

How does spending on x_i ($E_i = p_i x_i$) changes with change in p_i ?

Ordinary good:

- 1 Elastic demand, $e_{p_i}^i < -1$: 1% Δ in price $\Rightarrow >$ 1% Δ in demand, $\frac{\Delta E_i}{\Delta p_i} < 0$
- 2 Inelastic demand, $e_{p_i}^i > -1$: 1% Δ in price $\Rightarrow <$ 1% Δ in demand, $\frac{\Delta E_i}{\Delta p_i} > 0$
- 3 Unitary elastic demand, $e_{p_i}^i = -1$: 1% Δ in price $\Rightarrow$ exactly 1% Δ in demand, no change in E_i

Determinants of Elasticity of Demand Curve

Good's necessity

- If a good is necessary $\rightarrow$ its demand more likely to be inelastic.

Substitutability

- If a good can be substituted by different goods $\rightarrow$ its demand more likely to be elastic.

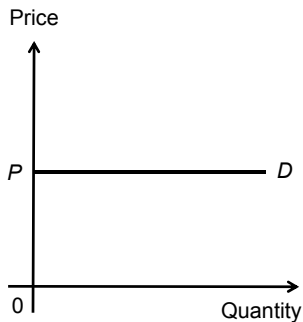
Time horizon

- Demand tends to be more elastic in longer time horizon.

Perfectly Elastic Demand Curve

$$|e_{p_i}^i| = \infty$$

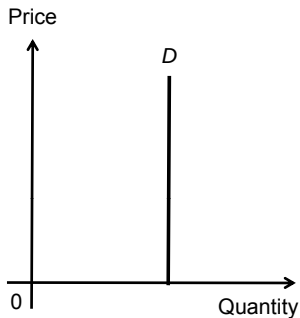
- If price $> p \Rightarrow q = 0$
- If price $= p \Rightarrow q \in [0, \infty]$



Perfectly Inelastic Demand Curve

$$|e_{p_i}^i| = 0$$

Zero quantity response to any price change.

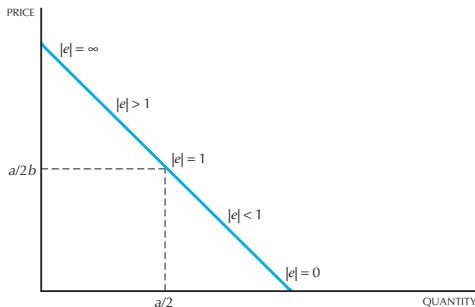


Price Elasticity along a Linear Demand Curve

Linear demand: $D_i(p_i) = a - bp_i$, $a, b > 0$

$$e_{p_i}^i = \frac{\frac{\partial D_i(p, m)}{\partial p_i}}{\frac{D_i(p, m)}{p_i}} = \frac{-b}{\frac{a - bp_i}{p_i}} = \frac{-bp_i}{a - bp_i}$$

Elasticity varies as we move along a demand curve.



Cross Price Elasticity of Demand

$$e_{p_j}^i = \frac{\% \Delta x_i}{\% \Delta p_j} = \frac{\frac{\partial D_i(p, m)}{\partial p_j}}{\frac{D_i(p, m)}{p_j}}$$

- 1 x_i is a gross complement to x_j : $e_{p_j}^i < 0$
- 2 x_i is a gross substitute to x_j : $e_{p_j}^i > 0$

Price Elasticity of Demand: Examples

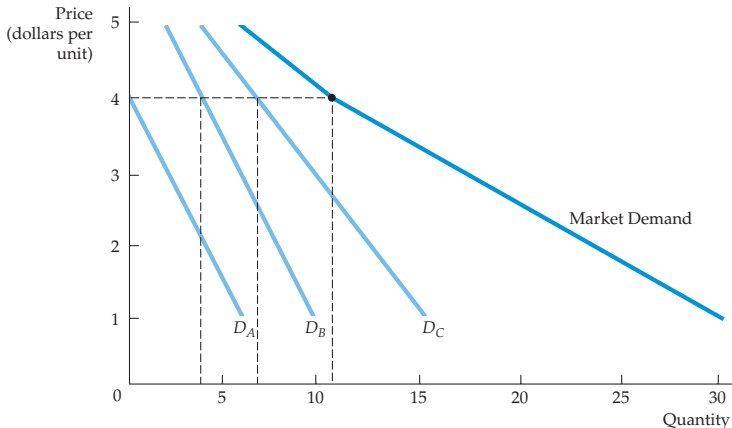
Janda, Mikolasek and Netuka (2010): Complete almost ideal demand system approach to the Czech alcohol demand. *Agriculture Economics*, 56(9), 421–434.

- Data from the Czech Budget Household Survey, around 11,000 observations
- Own price elasticities: beer -0.97 , wine -1.09 and spirits -1.21
- Cross-price elasticities for beer: change in price of wine -0.1143 , change in price of spirits 0.2047
- Cross-price elasticities for wine: change in price of beer -0.068 , change in price of spirits 0.2302

Dybczak, Toth and Vonka (2010): Effects of Price Shocks to Consumer Demand. Estimating the QUAIDS Demand System on Czech Household Budget Survey Data. Czech National Bank, Working Paper No. 8.

- similar data as above
- Own price elasticities: food -0.68 , clothing -0.49 , energy -0.96 , furniture -0.50 , health -0.722 , transportation -1.0 , education -0.34
- Cross-price elasticities: demand for food with change in price of clothing -0.257 , of transportation -0.183

Market Demand: summation of individual demands



Market demand curve is obtained by summing individual demand curves.

- Market demand curve shifts to the right as more consumers enter the market.
- Factors influence individual demands will also affect market demand.